Madi (Madeline) Abio

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Education

Griffith University Gold Coast, QLD
Bachelor of Computer Science / Bachelor of Electrical Engineering (Honours) *Mar 2022 – Nov 2026*

- Cumulative GPA: 6.32/7.

University of Utah Salt Lake City, Utah, USA
International Trimester Exchange *2024*

Awards & Scholarships

- **Academic Excellence** | *Griffith University, 2023–2025*
- **Chancellor's Scholarship** | *Griffith University, 2025*
- **Amplify Connector** | *Energy Queensland, 2025–Present*
- **Future of Energy Scholarship** | *Energy Queensland, 2023–2025*
- **Students in Power Bursary** | *Australian Institute of Power, 2022–2023*
- **Brighter Futures Scholarship** | *The Abedian Foundation, 2022*

Work Experience

Founding Engineer Aug 2025 – Present
VPTech *Gold Coast, QLD*

- Own end-to-end feature delivery across web applications, backend services, AI agents, and cloud infrastructure for an AI-native clinical platform.
- Built the company's first AI agent, a proof-of-concept that validated the product's flagship feature and set the architectural direction for the agent platform.
- Own code quality and maintainability, building and maintaining automated CI checks covering linting, security scanning, architectural boundaries, and design-system consistency.
- Mentor engineers in code cleanliness and engineering best practices through code review and team-wide standards.
- Drive technology discovery by scouting and evaluating emerging tools for adoption.
- Lead sprint planning.
- Improved developer experience by introducing stacked PRs, on-demand preview environments, and observability tooling for debugging agent behaviour.

Grid Technology Undergraduate Engineer Jan 2025 – Sep 2025
Energy Queensland *Brisbane, QLD*

- Developed an anomaly detection machine learning model to detect neutral faults targeting deployment across 10,000+ smart meters (see: Neutral Fault Detection project).
- Streamlined daily operational data analysis by building a solar panel data dashboard in Python/Power BI, reducing manual processing time from 15 minutes to 15 seconds.
- Managed project delivery using Agile principles, including facilitating weekly sprints.
- Automated various non-technical workflows.

Undergraduate Research Assistant Mar 2024 – Jun 2024
Griffith University *Gold Coast, QLD*

- Performed data pre-processing for a machine learning research project by cleaning and structuring large datasets using Pandas (Python), Bash, and the HDF5 format.

Cyber Security Platforms Intern Nov 2023 – Feb 2024
Energy Queensland *Brisbane, QLD*

- Developed a strong foundation in computer networking through self-directed training and mentorship, covering the OSI model, network protocols, and enterprise network architecture.
- Resolved ServiceNow tickets by applying network and security protocol knowledge.
- Gained exposure to enterprise cybersecurity practices, including cloud security, firewall management (Panorama), and Netskope.

Projects

Automated First-Order Logic Theorem Prover Rust, Apr 2026 – May 2026

- Implemented a backward sequent-calculus (LK′) proof search engine for first-order logic in Rust.
- Extended it with a 6-class priority schedule ordering rule applications by productivity, and an iterative-deepening envelope around the depth-first kernel.
- Built a parallelized engine sweep and benchmark generator in Python.

Full-Stack LeetCode Web App (madiab.io) TypeScript, Mar 2026

- Designed a full-stack web app hosting my LeetCode solve times and portfolio.
- NestJS backend, Next.js frontend, PostgreSQL database, Prisma ORM, and Better Auth for authentication.
- Linked the projects section to my GitHub repositories via the public API.
- Hosted on Railway.

Digital Bass Synthesizer on Microcontroller Embedded C, Sep 2025 – Oct 2025

- Developed bare-metal C firmware to generate waveforms, control ADSR envelopes, and handle input from potentiometers (via ADC) and a matrix keypad.
- Engineered a custom external clock circuit to bridge the I^2C peripheral for compatibility with the I^2S DAC.
- Drove the 240x320 LCD over SPI for waveform and real-time ADSR visualization.
- Implemented DMA ping-pong buffers for continuous audio streaming and optimized ISR timing for stable waveform output.

Neutral Fault Detection (Anomaly Detection) Python, Feb 2025 – Sep 2025

- Developed an in-house neutral fault detection algorithm to investigate excessive false positives in a third-party solution.
- Identified systematic false positives on solar PV sites, caused by violated voltage–current modelling assumptions making normal solar behaviour resemble fault signatures.
- Proposed a revised approach that explicitly accounted for solar PV electrical behaviour.
- Designed and prototyped a scalable architecture targeting deployment across 10,000+ smart meters.
- Built a parallelized feature extraction and evaluation pipeline with time-ordered, meter-isolated validation and failure-mode analysis.

Encryption Client C++, May 2025

- Developed a terminal-based multi-round encryption/decryption client, applying OOP principles (abstract base classes, inheritance, polymorphism) for modular, reusable design.
- Implemented a finite-state-machine menu with input validation and exception handling, using the STL and lambdas for efficient message processing.

Clubs & Societies

- **Ravi's Study Program** | *Mentee, 2026*
- **Griffith University Coding Club (GCC)** | *ICPC Div B Competitor & Member, 2025–Present*
- **UQ Computing Society** | *Lead Guitar Player (UQCS Band), Competitive Programmer, 2025–Present*
- **UQ Fintech Society** | *Member, 2025–Present*
- **Griffith University Advanced Robotics Development (GUARD)** | *Member, 2023–Present*
- **Griffith University Gold Coast IEEE Student Branch** | *Administrator, 2023*

Skills

- **Programming Languages**: Bash, C, C++, C#, Python, Rust, TypeScript
- **Software Development & Tools**: Agentic Software Development (with Claude Code), AWS, CI/CD, Drizzle, Fastify, Git, Linux, NestJS, Next.js, Object-Oriented Programming, oRPC, PostgreSQL, Prisma, React, Test-Driven Development
- **Machine Learning & Data Science**: Applied Machine Learning, Data Science, h5py, NumPy, Pandas, scikit-learn
- **LLMOps**: AgentCore, LangChain, LangGraph, LLM orchestration
- **Embedded & FPGA**: Embedded/Bare-Metal C, PCB Design, SystemVerilog
- **Professional Skills**: Communication, Problem Solving, Project Management, Teamwork, Technical Writing